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Apparatus comprising a bias current generator

Abstract

An apparatus comprising a bias current generator for generating a bias current, the bias current generator comprising two or more negative threshold transistors of the same type and coupled in series.

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Background/Summary

(1) The present disclosure relates to an apparatus comprising a bias current generator. In particular, the present disclosure relates to a bias current generator comprising a plurality of negative threshold transistors.

BACKGROUND

(2) Bias current generators are circuits used to generate a bias current. A bias current typically refers to a current that is generated and provided to other circuitry, for example on a chip, that requires the bias current for its operation. Multiple nanoampere bias current generators, which are used to generate bias currents on the order of nanoamps, are required in low power chips. Trimming of such circuits using prior art is very time-consuming and hence costly.

(3) Circuits which may include multiple nanoampere bias current generators (for example, eight to ten nanoampere bias current generators) include but are not limited to: a bandgap cell; a low frequency oscillator; a power on reset circuit; an undervoltage-lockout (UVLO) circuit; a test mux buffer; a low dropout regulator (LDO), a LDO current limit bias circuit; and a low voltage start-up circuit.

(4) FIG. 1 is a schematic of a known bias current generator **100** comprising a negative threshold NMOS device **102** and a poly resistor **104** for generating a bias current I_{bias} . The bias current I_{bias} is as follows:

$$(5) \quad I_{bias} = \frac{V_{th_{NMOS}}}{R_{poly}} \quad (1)$$

where $V_{th.sub.NMOS}$ is the threshold voltage of the device **102** and $R_{sub.poly}$ is the resistance of the poly resistor.

(6) The disadvantages of the generator **100** are as follows: Such circuits include a huge resistor meaning that the circuit will occupy a significant area on a chip compared to other analog active circuits. This is particularly relevant when multiple bias current generators are required on a single chip. Multiple measurements are required to evaluate the bias current I_{bias} as exemplified by

equation (1). Specifically, both $V_{th.sub.NMOS}$ and $R_{sub.poly}$ must be measured. or the generated current itself must be measured. However, to measure the generated bias current directly it may be necessary to scale up the bias current to a level that is measurable. This procedure can introduce scaling errors. The poly resistor **104** will only carry a small current in nanoampere applications and its resistance $R_{sub.poly}$ will be large, meaning measurement of the resistance $R_{sub.poly}$, and therefore the bias current I_{bias} , can be time consuming and can result in an increase in cost of measurement. Although possible to scale the poly resistor **104** down to decrease its resistance $R_{sub.poly}$ to speed up the measurement process, there will be a scaling error that can result in the inaccuracy of measurement. The scaled-up version of generated I_{bias} current can be measured to speed up the process but, as discussed previously, it will also have a scaling error. The challenge in measuring the bias current I_{bias} can result in a more challenging trimming process than is desirable.

SUMMARY

(7) It is desirable to provide a bias current generator that occupies a smaller area on a chip than known systems.

(8) Furthermore, it is desirable to provide a bias current generator that enables a measurement of its bias current that is faster, more accurate and/or less costly than known systems.

(9) Furthermore, it is desirable to provide a bias current generator that provides an improved trimming process compared with known systems.

(10) According to a first aspect of the disclosure there is provided an apparatus comprising a bias current generator for generating a bias current, the bias current generator comprising two or more negative threshold transistors of the same type and coupled in series.

(11) Optionally, the two or more negative threshold transistors are NMOS or PMOS.

(12) Optionally, the bias current generator comprises a current mirror configured to output the bias current.

(13) Optionally, the current mirror comprises a first mirror transistor, and a second mirror transistor, wherein the first mirror transistor is configured to receive the bias current and to mirror the bias current to the second mirror transistor, and the second mirror transistor is configured to output the bias current as received from the first mirror transistor.

(14) Optionally, the two or more negative threshold transistors comprise a first negative threshold transistor, and one or more second negative threshold transistors.

(15) Optionally, at least one of the one or more second negative threshold transistors has its gate coupled to a second voltage, its drain terminal, or its source terminal.

(16) Optionally, the bias current generator comprises a transistor set comprising the one or more second negative threshold transistors, the first negative threshold transistor comprising a first terminal coupled to a first terminal of the transistor set, and a gate terminal coupled to a second terminal of the transistor set.

(17) Optionally, the first negative threshold transistor comprises a second terminal coupled to a first voltage and/or the second terminal of transistor set is coupled to a second voltage.

(18) Optionally, the bias current generator comprises a cascode transistor, the second terminal of the first negative threshold transistor being coupled to the first voltage via a cascode transistor.

(19) Optionally, the bias current generator comprises a current mirror configured to output the bias current.

(20) Optionally, the current mirror comprises a first mirror transistor, and a second mirror transistor, wherein the second terminal of the first negative threshold transistor is coupled to the first voltage via the first mirror transistor, the first mirror transistor is configured to receive the bias current and to mirror the bias current to the second mirror transistor, and the second mirror transistor is configured to output the bias current as received from the first mirror transistor.

(21) Optionally, the bias current generator comprises a cascode transistor, the second terminal of the first negative threshold transistor being coupled to the first voltage via the first mirror transistor

and the cascode transistor.

(22) Optionally, the apparatus comprises measurement circuitry configured to measure a first quantity that is dependent on the bias current.

(23) Optionally, the first quantity comprises a threshold voltage, the threshold voltage being dependent on the bias current.

(24) Optionally, the measurement circuitry comprises a buffer circuit coupled to a threshold voltage terminal and configured to output the threshold voltage.

(25) Optionally, the first negative threshold transistor has a width of approximately 250 nm to approximately 500 nm and a length of approximately 20 micrometers and/or the one or more second negative threshold transistors have a width of approximately 500 nm and a length of approximately 20 micrometers.

(26) Optionally, the apparatus comprises trimming circuitry configured to provide a trimming signal to the bias current generator, the bias current generator adjusting the bias current in response to the trimming signal to provide a desired bias current.

(27) Optionally, the apparatus comprises measurement circuitry configured to measure a first quantity that is dependent on the bias current, wherein the trimming circuitry is configured to receive the measurement of the first quantity, determine the trimming signal necessary to provide to the bias current generator such that the bias current generator will adjust the bias current to provide the desired bias current, and provide the determined trimming signal to the bias current generator.

(28) Optionally, the first quantity comprises a threshold voltage, the threshold voltage being dependent on the bias current.

(29) Optionally, the measurement circuitry comprises a buffer circuit coupled to a threshold voltage terminal and configured to output the threshold voltage.

(30) Optionally, the trimming circuitry comprises a memory element configured to store a look up table comprising a plurality of trimming signal settings, each trimming signal setting being associated with a value of the first quantity.

(31) Optionally, the trimming circuitry is configured to determine the trimming signal by selecting, from the look up table, the trimming signal setting associated with the value of the first quantity as received by the trimming circuitry, and setting the trimming signal based on the trimming signal setting as selected, thereby determining the determined trimming signal.

(32) Optionally, the first quantity comprises a threshold voltage, the threshold voltage being dependent on the bias current, such that the value of the first quantity as received by the trimming circuit is the value of the measured threshold voltage.

(33) Optionally, the measurement circuitry comprises a buffer circuit coupled to a threshold voltage terminal and configured to output the threshold voltage.

(34) Optionally, the two or more negative threshold transistors comprises a first negative threshold transistor, and a plurality of second negative threshold transistors, the bias current generator comprises a transistor set comprising a plurality of bit portions, each bit portion comprising at least one of the second negative threshold transistors, wherein the trimming signal is configured to set each bit portion in an enabled or disabled state.

(35) Optionally, the trimming circuitry comprises a memory element configured to store a look up table comprising a plurality of trimming signal settings, each trimming signal setting being associated with a value of the first quantity.

(36) Optionally, the trimming circuitry is configured to determine the trimming signal by selecting, from the look up table, the trimming signal setting associated with the value of the first quantity as received by the trimming circuitry, and setting the trimming signal based on the trimming signal setting as selected, thereby determining the determined trimming signal.

(37) Optionally, the first quantity comprises a threshold voltage, the threshold voltage being dependent on the bias current, such that the value of the first quantity as received by the trimming circuit is the value of the measured threshold voltage.

- (38) Optionally, the measurement circuitry comprises a buffer circuit coupled to a threshold voltage terminal and configured to output the threshold voltage.
- (39) According to a second aspect of the disclosure there is provided a method of providing a bias current comprising providing a bias current generator comprising two or more negative threshold transistors of the same type and coupled in series.
- (40) Optionally, the method comprises outputting the bias current using a current mirror.
- (41) Optionally, the method comprises providing a trimming signal to the bias current generator using trimming circuitry, and adjusting the bias current in response to the trimming signal to provide a desired bias current.
- (42) Optionally, the method comprises measuring a first quantity that is dependent on the bias current using measurement circuitry, receiving the measurement of the first quantity using the trimming circuitry, determining the trimming signal necessary to provide to the bias current generator such that the bias current generator will adjust the bias current to provide the desired bias current, and providing the determined trimming signal to the bias current generator.
- (43) Optionally, the first quantity comprises a threshold voltage.
- (44) Optionally, the trimming circuitry comprises a memory element configured to store a look up table comprising a plurality of trim signal settings, each trimming signal setting being associated with a value of the first quantity, the method comprising determining the trimming signal by selecting, from the look up table, the trimming signal setting associated with the value of the first quantity as received by the trimming circuitry, and setting the trimming signal based on the trimming signal setting as selected, thereby determining the determined trimming signal.
- (45) Optionally, the two or more negative threshold transistors comprises a first negative threshold transistor and a plurality of second negative threshold transistors, the bias current generator comprises a transistor set comprising a plurality of bit portions, each bit portion comprising at least one second negative threshold transistor, the method comprising setting each bit portion in an enabled or disabled state using the trimming signal, thereby adjusting the bias current in response to the trimming signal to provide a desired bias current.
- (46) It will be appreciated that the method of the second aspect may include providing and/or using features set out in the first aspect and can incorporate other features as described herein.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosure is described in further detail below by way of example and with reference to the accompanying drawings, in which:
- (2) FIG. 1 is a schematic of a known bias current generator;
- (3) FIG. 2(a) is a schematic of an apparatus comprising a bias current generator in accordance with a first embodiment of the present disclosure, FIG. 2(b) is a schematic of a specific embodiment of the bias current generator, FIG. 2(c) is a schematic of a further specific embodiment of the bias current generator;
- (4) FIG. 3(a) is a schematic of an apparatus comprising a bias current generator in accordance with a second embodiment of the present disclosure, FIG. 3(b) is a schematic of the apparatus having a specific embodiment of the current mirror;
- (5) FIG. 4(a) is a schematic of an apparatus comprising a bias current generator in accordance with a third embodiment of the present disclosure, FIG. 4(b) is an alternative embodiment of the apparatus of FIG. 4(a), FIG. 4(c) is a further alternative embodiment of the apparatus of FIG. 4(a);
- (6) FIG. 5(a) is a schematic of an apparatus in accordance with a fourth embodiment of the present disclosure, FIG. 5(b) is a schematic of an alternative embodiment of the apparatus of FIG. 5(a);
- (7) FIG. 6(a) is a schematic of an apparatus comprising the bias current generator in accordance

with a fifth embodiment of the present disclosure, FIG. 6(b) is a schematic of the apparatus showing a specific embodiment of the measurement circuitry, FIG. 6(c) is a schematic of the apparatus showing a further specific embodiment of the measurement circuitry;

(8) FIG. 7(a) is a schematic of an apparatus comprising the bias current generator in accordance with a sixth embodiment of the present disclosure, FIG. 7(b) is a schematic of the apparatus comprising the bias current generator in accordance with a seventh embodiment of the present disclosure, FIG. 7(c) is a schematic of the apparatus having a specific measurement circuitry and trimming circuitry arrangement, FIG. 7(d) is a schematic of an apparatus in accordance with an eighth embodiment of the present disclosure;

(9) FIG. 8(a) is a schematic of an apparatus comprising a bias current generator in accordance with a ninth embodiment of the present disclosure, FIG. 8(b) is a schematic of illustrating the relative size of the second negative threshold compared with the poly resistor;

(10) FIG. 9(a) is a schematic of an apparatus comprising a bias current generator in accordance with a tenth embodiment of the present disclosure, FIG. 9(b) is a schematic of an apparatus comprising a bias current generator in accordance with an eleventh embodiment of the present disclosure;

(11) FIG. 10 is a graph showing a relationship between the threshold voltage and the bias current for an embodiment of a bias current generator;

(12) FIG. 11(a) is a schematic of an apparatus comprising a bias current generator in accordance with a twelfth embodiment of the present disclosure, FIG. 11(b) is a schematic of the apparatus showing an alternative embodiment with the gate coupling of the second negative threshold transistors differing from as shown in FIG. 11(a), FIG. 11(c) is a table showing an example look up table; and

(13) FIG. 12(a) is a table showing example simulation results for an implementation of the bias current generator of the present disclosure, FIG. 12(b) is a table showing simulation results for an implementation of the circuit of FIG. 1.

DETAILED DESCRIPTION

(14) FIG. 2(a) is a schematic of an apparatus **200** comprising a bias current generator **202** for generating a bias current I_{bias} in accordance with a first embodiment of the present disclosure. The bias current generator **202** comprises two negative threshold transistors of the same type that are coupled in series. In further embodiments, the bias current generator **202** may comprise more than two negative threshold transistors. For example, PMOS and NMOS are transistor “types”.

(15) A negative threshold transistor is a transistor that will conduct a current when a voltage of 0V is applied across its gate and source.

(16) The two negative threshold transistors may comprise a first negative threshold transistor **204** and a second negative threshold transistor **206**. In further embodiments, there may be two or more second negative threshold transistors.

(17) In comparison with the generator **100** of FIG. 1, the present embodiment replaces the poly resistor **104** with the second negative threshold transistor **206**. Further embodiments may replace the poly resistor **104** with two or more second negative threshold transistors. The removal of the poly resistor **104** can result in a decrease in the bias current generator **200** size when compared with the generator **100**. Therefore, embodiments of the present disclosure can provide a bias current generator that can occupy a smaller area than known systems.

(18) The second negative threshold transistor **206** provides a resistance thereby delivering the functionality as provided by the poly resistor **104**. A plurality of second negative threshold transistors coupled in series provides a resistance, with an increasing number of second negative threshold transistors in the chain providing a higher resistance value.

(19) The one or more second negative threshold transistors in series may be referred to as a resistor device, which is descriptive of their function in providing a resistance.

(20) In a specific embodiment, the first negative threshold transistor **204** may have a width of

approximately 250 nm to approximately 500 nm and a length of approximately 20 micrometers. Additionally, or alternatively, the second negative threshold transistors **206** may have a width of approximately 500 nm and a length of approximately 20 micrometers. In further embodiments where there are two or more second threshold transistors, one or more of these second negative threshold transistors may have the dimensions as discussed previously. Multiple long channel and narrow width second negative threshold transistors, for example having the dimensions as discussed here, may be coupled in series to provide a high resistance, which may be used for nA bias current generation.

(21) The bias current generators of the present disclosure may exhibit PVT (process, supply, temperature) variation; may require no startup; and may provide a supply independent bias current.

(22) FIG. 2(b) is a schematic of a specific embodiment of the bias current generator **200** where the negative threshold transistors **204**, **206** are n-type transistors, for example NMOS transistors.

(23) FIG. 2(c) is a schematic of a specific embodiment of the bias current generator **200** where the negative threshold transistors **204**, **206** are p-type transistors, for example PMOS transistors.

(24) FIG. 3(a) is a schematic of an apparatus **300** comprising a bias current generator **302** for generating a bias current I_{bias} in accordance with a second embodiment of the present disclosure. The bias current generator **302** is as described for the bias current generator **202**. However, in the present embodiment, the bias current generator **302** further comprises a current mirror **304** configured to output the bias current I_{bias} . The current mirror **304** may be used to distribute the bias current I_{bias} to other circuits.

(25) In the embodiments described herein, there may be illustrated a 1:1 mirroring of the bias current I_{bias} . However, it will be appreciated that in further embodiments the bias current I_{bias} may be scaled up or down as part of the mirroring process in accordance with the understanding of the skilled person.

(26) It will be appreciated that the embodiments described herein are illustrated using NMOS transistors. Alternatively other transistors types, such as PMOS may be used

(27) FIG. 3(b) is a schematic of the apparatus **300** having a specific embodiment of the current mirror **304**. In the present embodiment the current mirror comprises mirror transistors **306**, **308**. In operation, the mirror transistor **306** receives the bias current I_{bias} and mirrors the bias current I_{bias} to the mirror transistor **308**. The mirror transistor **308** then outputs the bias current I_{bias} .

(28) FIG. 4(a) is a schematic of an apparatus **400** comprising a bias current generator **402** for generating a bias current I_{bias} in accordance with a third embodiment of the present disclosure. In the present embodiment there is a further second negative threshold transistor **404** when compared with the previously discussed embodiments. In the present embodiment the second negative threshold transistors have their gates coupled to a voltage v_2 .

(29) FIG. 4(b) is an alternative embodiment of the apparatus **400**, where the gates of the second negative threshold transistors **206**, **404** are coupled to their source terminals. FIG. 4(c) is an alternative embodiment of the apparatus **400**, where the gates of the second negative threshold transistors **206**, **404** are coupled to their drain terminals.

(30) The second negative threshold transistors **206**, **404** provide a resistance, with the different gate coupling combinations presented in FIGS. 4(a)-(c) providing different resistances, due to the different gate couplings. The sensitivity of the resistance due to process and/or temperature variations will also vary depending on the gate coupling configuration of the negative threshold transistors **206**, **404**.

(31) The transistors **206**, **404** may form a transistor set **406** such that a first terminal **408** of the first negative threshold transistor **204** is coupled to a first terminal **410** of the transistor set **406** and a gate terminal **412** of the first negative threshold transistor **204** is coupled to a second terminal **414** of the transistor set **406**. In the embodiments shown in FIGS. 4(a)-(c) the first negative threshold transistor **204** comprises a second terminal **416** that is coupled to a voltage v_1 and the second terminal **414** of the transistor set **406** is coupled to the voltage v_2 . For clarity, the reference

numerals have only been applied to FIG. 4(a), however it will be evident to the skilled person that this configuration is also present in FIG. 4(b) and FIG. 4(c) and some of the other embodiments as disclosed herein.

(32) FIG. 5(a) is a schematic of an apparatus **500** comprising a bias current generator **502** for generating a bias current I_{bias} in accordance with a fourth embodiment of the present disclosure. In the present embodiment, the bias current generator **502** further comprises a cascode transistor **504**. The terminal **416** is coupled to the voltage v_1 via the cascode transistor **504**.

(33) The cascode transistor **504** is configured to protect the first negative threshold transistor **204** from variations in the supply voltage (in the present embodiment represented by v_1) using the cascode technique. The cascode transistor **504** and the first negative threshold transistor **204** may be of the same type, which in the present example is NMOS. FIG. 5(b) is a schematic of an alternative embodiment of the apparatus **500** further comprising the current mirror **304**.

(34) It will be appreciated that in FIGS. 5(a)-(b), and in any other embodiments disclosed herein, the gate coupling of the negative threshold transistors **206**, **404** may be as shown in one of FIGS. 4(a)-(c) or in any other suitable configuration in accordance with the understanding of the skilled person.

(35) FIG. 6(a) is a schematic of an apparatus **600** comprising the bias current generator **202** for generating a bias current I_{bias} in accordance with a fifth embodiment of the present disclosure. The apparatus **600** comprises measurement circuitry **602** configured to measure a first quantity that is dependent on the bias current I_{bias} .

(36) The first quantity may, for example, comprise a threshold voltage V_{th} , where the threshold voltage V_{th} is dependent on the bias current I_{bias} . The threshold voltage V_{th} may be the threshold voltage V_{th} of any of the negative threshold transistors of the bias current generator **202**.

(37) FIG. 6(b) is a schematic of the apparatus **600**, showing a specific embodiment where the measurement circuitry **602** is coupled to the terminal between the first and second negative threshold transistors **204**, **206** to measure the threshold voltage V_{th} of the first negative threshold transistor **204**.

(38) Measurement of the threshold voltage V_{th} provides a faster and more accurate method of measuring the bias current I_{bias} than as provided by the system of FIG. 1. Specifically, there is a relationship between the bias current I_{bias} and the threshold voltage V_{th} in the bias current generators as disclosed herein that means that the bias current I_{bias} can be determined by a threshold voltage V_{th} measurement, and without additional measurements. This is not possible in the system of FIG. 1, as the use of the poly resistor **104** means that the resistance of the poly resistor **104** must also be measured to evaluate the bias current I_{bias} .

(39) In summary, the bias current I_{bias} can be measured simply by measuring the threshold voltage V_{th} of one of the negative threshold transistors, thereby providing a faster method of current measurement than is possible in known systems. This is because two negative threshold transistors of the same type coupled in series will track each other over process and temperature variations. Furthermore, bias current I_{bias} is proportional to threshold voltage V_{th} in circuits having this configuration.

(40) FIG. 6(c) is a schematic of the apparatus **600** showing a specific embodiment of the measurement circuitry **602**. In the present embodiment, the measurement circuitry comprises a buffer circuit **604** coupled to the threshold voltage terminal and configured to output the threshold voltage V_{th} . The buffer circuit **604** comprises an op amp **606** in the present embodiment. A buffer circuit **604** is usually very fast settling.

(41) FIG. 7(a) is a schematic of an apparatus **700** comprising the bias current generator **202** for generating a bias current i_{bias} in accordance with a sixth embodiment of the present disclosure. In the present embodiment, the apparatus **700** comprises trimming circuitry **702** configured to provide a trimming signal **704** to the bias current generator **202**. The bias current generator **202** is configured to adjust the bias current I_{bias} in response to the trimming signal **704** to provide a

desired bias current I_{bias} .

(42) The trimming circuitry **702** is arranged to provide a trimming function. Trimming is a well-known procedure in the technical field, where a circuit may provide an output that is not as intended, and an aspect of the circuit is adjusted to provide the desired output. In the present area of bias current generation, a bias current generator may generate a bias current that does not have the value as intended, for example due to process or temperature variations. The trimming procedure acts to adjust one or more aspects the bias current generator until the bias current value is as desired.

(43) FIG. 7(b) is a schematic of the apparatus **700** comprising the bias current generator **202** for generating a bias current I_{bias} in accordance with a seventh embodiment of the present disclosure. In the present embodiment, and compared with the apparatus shown in FIG. 7(a), the apparatus **700** further comprises the measurement circuitry **602**.

(44) In operation, the measurement circuitry **602** measures the first quantity, which is then provided to the trimming circuitry **702**. The trimming circuitry **704** determines the suitable trimming signal **704** necessary to provide to the bias current generator **202** such that the bias current generator **202** will adjust the bias current I_{bias} to provide the desired bias current I_{bias} . The trimming signal **704** as determined is then provided to the bias current generator **202**.

(45) FIG. 7(c) is a schematic of the apparatus **700** having a specific measurement circuitry **602** and trimming circuitry **702** arrangement. In the present embodiment, the first quantity comprises the threshold voltage V_{th} which may be measured as described previously for FIGS. 6(b) and FIG. 6(c).

(46) The trimming of the bias current generator can be carried out in production more reliably and quicker than known methods due to the increased speed with which the I_{bias} current can be determined using the methods and tools of the present disclosure.

(47) The trimming circuitry **702** may comprise a memory element **706** configured to store a look up table comprising a plurality of trimming signal settings, where each trimming signal setting is associated with a value of the first quantity. As bias current can be related to the first quantity (for example, where it comprises the threshold voltage V_{th}), it is possible to trim using the lookup table.

(48) In a specific embodiment, the trimming circuitry **702** may be configured to determine the trimming signal **704** by selecting, from the look up table, the trimming signal setting associated with the value of the first quantity as received by the trimming circuitry **702** and setting the trimming signal **704** based on the trimming signal setting as selected.

(49) As discussed previously, the first quantity may comprise the threshold voltage V_{th} , such that the value of the first quantity as received by the trimming circuit **702** is the value of the measured threshold voltage V_{th} .

(50) FIG. 7(d) is a schematic of an apparatus **708** in accordance with an eighth embodiment of the present disclosure. Compared with the apparatus **700** of FIG. 6(c), in the present embodiment, the apparatus comprises a bias current generator **710** having two second negative threshold transistors **206**, **404**. The bias current generator **710** comprises the transistor set **406** which comprises bit portions **712**, **714**, with each bit portion **712**, **714** comprising one of the second negative threshold transistors **206**, **404**. The trimming signal **704** is configured to set each bit portion in an enabled or disabled state to adjust the bias current provided by the bias current generator **710**.

(51) Enabled may refer to the normal operation of the second negative threshold transistor of the bit portion, where its resistance contributes to the bias current generation. Disabled may refer to the second negative threshold transistor being bypassed. This functionality may be provided by having a switch in parallel with each bit portion, such that in the enabled state the switch is open, such that current will flow through the transistor, and in the disabled state the switch is closed such that current will flow through the switch, and not the transistor.

(52) Although FIGS. 6(a)-(c) and FIGS. 7(a)-(c) are shown for the bias current generator **202** of

FIG. 2(b), it will be appreciated that further embodiments may alternatively comprise any of the other bias current generators as disclosed herein, or any combination of features of bias current generators as disclosed herein, in accordance with the understanding of the skilled person.

(53) FIG. 8(a) is a schematic of an apparatus **800** comprising a bias current generator **802** for generating a bias current I_{bias} in accordance with a ninth embodiment of the present disclosure. In the present embodiment, the negative threshold transistors **204**, **206** are NMOS type, however it will be appreciated that in further embodiments they may, for example, be PMOS type.

(54) FIG. 8(b) is a schematic of illustrating the relative size a practical implementation of the second negative threshold transistor **206** compared with a practical implementation of the poly resistor **104**. In this example, the second negative threshold transistor **206** occupies approximately nine times less area than the poly resistor **104**.

(55) FIG. 9(a) is a schematic of an apparatus **900** comprising a bias current generator **902** for generating a bias current I_{bias} in accordance with a tenth embodiment of the present disclosure. In the present embodiment, the bias current generator **902** comprises second negative threshold transistors **206**, **404**, **904**, **906** having their gates coupled to the voltage V_{SS_A} .

(56) FIG. 9(b) is a schematic of an apparatus **908** comprising a bias current generator **910** for generating a bias current I_{bias} in accordance with an eleventh embodiment of the present disclosure. In the present embodiment, the bias current generator **910** comprises second negative threshold transistors **206**, **404**, **904**, **906** having their gates coupled to their drains.

(57) FIG. 10 is a graph **1000** showing a relationship between the threshold voltage V_{th} and the bias current I_{bias} for a practical implementation of an embodiment of a bias current generator as disclosed herein. It can be observed that bias current I_{bias} is proportional to the threshold voltage V_{th} . Therefore it is possible to determine the bias current I_{bias} from a measurement of the threshold voltage V_{th} .

(58) The x-axis (millivolts) is the threshold of the first negative threshold transistor **204**, with the y-axis (nanoamperes) being the bias current I_{bias} flowing through the transistor **204**. Such a relationship may be derived from simulation and/or experimental measurement, in accordance with the understanding of the skilled person. This may then be used generate a suitable look up table for providing the necessary trimming functionality.

(59) FIG. 11(a) is a schematic of an apparatus **1100** comprising a bias current generator **1102** for generating a bias current I_{bias} in accordance with a twelfth embodiment of the present disclosure. The present embodiment includes circuitry for trimming. Specifically, there are includes switches **1104**, **1106** in parallel with bit portions **712**, **714**. This circuitry provides two-bit trimming. By shorting the series connected second negative threshold transistors of a bit portion, it is possible to change the total resistance of the transistor set **406**, thereby altering the bias current I_{bias} . The measurement circuitry **602** and trimming circuitry **702** have been omitted from this drawing to aid clarity of the schematic. However, it will be appreciated that they are used in this embodiment to measure the threshold voltage V_{th} and then provide the required trimming functionality.

(60) The switches **1104**, **1106** may, for example, be implemented using PMOS devices or transmission gates.

(61) FIG. 11(b) is a schematic of the apparatus **1100** showing an alternative embodiment with the gate coupling of the second negative threshold transistors differing from as shown in FIG. 11(a).

(62) FIG. 11(c) is a table **1103** showing an example look up table that may be stored in the memory element **706** of the trimming circuitry **702**. In operation, when a voltage threshold V_{th} in the lefthand column is measured, the trimming signal **704** is generated based on the corresponding trim code (in this example a 4-bit trim code) in the right hand column, to control the bit portions **712**, **714**, to provide the required resistance of the transistor set **406**, and therefore the trimmed bias current I_{bias} . This procedure may be undertaken during chip production. In the present example, the look up table shows threshold voltage V_{th} through the first negative threshold transistor **204** versus the bias current flowing through the first negative threshold transistor **204**.

(63) In summary, the use of bit portions **712**, **714** having parallel coupled switches **1104**, **1106** enables trimming by provides a mechanism for adding and/or removing the series coupled second negative threshold transistors **404**, **904**, **906**.

(64) Consider the transistor **906** as functions as a resistor R then the series coupling of transistors **404**, **904** provide a resistance of $2R$. If we include 4 such devices it will offers $4R$. In the present embodiment, there is shown a two bit binary array with the transistor **906** being the least significant bit (LSB).

(65) Instead of using one device as LSB It is possible to use a combination of devices to create LSB (where some of them have gate connected to vss and some of them are connected to drain or source).

(66) If we bring the buffered threshold voltage terminal out via test mux we can simply look at the threshold voltage V_{th} and predict what the bias current I_{bias} is with an approximate $\pm 10\%$ accuracy, as illustrated by FIG. **10**.

(67) It will be appreciated that in a practical implementation, trimming accuracy of $\pm 10\%$ is sufficient and therefore coarse trimming codes may be used. Final trimming accuracy will be limited by the mismatch as shown on FIG. **10**. For example, for a given measured voltage threshold V_{th} , a bias current of 1 nA could be ± 100 pA within target.

(68) In known systems it is not possible to trim a nA bias current in production economically because of the massive settling time related to measuring nA current and trim code sweep using the poly resistor **104**. Large trim time will lead to significant cost issues. This is resolved by the embodiments disclosed herein.

(69) The trimming methods disclosed herein do not require nA range current measurement, which cannot be done in production reliably and fast. In the embodiments disclosed herein it is possible to measure the threshold voltage V_{th} and use a lookup table to apply the trim function. This results in a faster process than known systems which require both a measurement of the poly resistor **104** resistance and the threshold voltage V_{th} .

(70) In summary, it is possible to trim a generated bias current I_{bias} in a time-efficient and hence low-cost manner by simply measuring the threshold voltage V_{th} of the first negative threshold transistor **204** and using a look-up table

(71) FIG. **12(a)** is a table **1200** showing example simulation results for a practical implementation of the bias current generator **800** of the present disclosure. FIG. **12(b)** is a table **1201** showing simulation results for a practical implementation of the circuit of FIG. **1**, using the poly resistor **104**. Shown on both tables is a bias current **1202** as measured for different temperature values **1204**.

(72) There was found to be 30% less temperature variation through the use of the second negative threshold device **206** than the poly resistor-based bias generator of the circuit **100**. Furthermore, the bias current generator **800** of the present disclosure can be fine-tuned to reduce variation further.

(73) Temperature coefficient of the bias current can be adjusted based on the size and dimensions of the negative threshold transistors and their gate connections, as discussed previously.

(74) The impact of the width (W) and length (L) of the negative threshold transistors is summarised below for practical implementations of the bias current generators as disclosed herein: $W/L=250$ nm/ $20\ \mu\text{m}$, gives $6\times$ PVT variation in current and $2.6\times$ temperature variation. W/L of 500 nm/ $20\ \mu\text{m}$ for first negative threshold transistor **204** and 500 nm/ $20\ \mu\text{m}$ for second negative threshold transistors gives $4\times$ PVT and $1.7\times$ temperature variation. We can fine-tune PVT or temperature spread depending on the requirements of a given application. Generally, a bigger device gives better PVT spread and temperature spread Above 500 nm micron width, returns were found to be diminishing. It is anticipated that other size combination will provide a performance in the range stated above.

(75) By reducing one or more of process, supply and temperature variations compared with known systems, there can be provided more reliable and repeatable bias current generation circuitry with

reduced variation in performance between different circuits.

(76) Common reference numerals or variables between Figures represent common features.

(77) Various improvements and modifications may be made to the above without departing from the scope of the disclosure.

Claims

1. An apparatus comprising: a bias current generator comprising two or more negative threshold transistors coupled in series, wherein the bias current generator is configured to generate a bias current; a measurement circuitry configured to measure a first quantity that is dependent on the bias current; and a trimming circuitry configured to: receive the measurement of the first quantity; based on the measurement of the first quantity, determine a trimming signal necessary to provide to the bias current generator such that the bias current generator will adjust the bias current to provide a desired bias current; and provide the trimming signal determined to the bias current generator, wherein the bias current generator adjusts the bias current in response to the trimming signal to provide the desired bias current.
2. The apparatus of claim 1, wherein the two or more negative threshold transistors are NMOS or PMOS.
3. The apparatus of claim 1, wherein the bias current generator comprises a current mirror configured to output the bias current.
4. The apparatus of claim 3, wherein the current mirror comprises: a first mirror transistor; and a second mirror transistor; wherein: the first mirror transistor is configured to receive the bias current and to mirror the bias current to the second mirror transistor; and the second mirror transistor is configured to output the bias current as received from the first mirror transistor.
5. The apparatus of claim 1, wherein the two or more negative threshold transistors comprise: a first negative threshold transistor; and one or more second negative threshold transistors.
6. The apparatus of claim 5, wherein a gate terminal of at least one of the one or more second negative threshold transistors is coupled to: a second voltage; a drain terminal of at least one of the one or more second negative threshold transistors; or a source terminal of at least one of the one or more second negative threshold transistors.
7. The apparatus of claim 5, wherein: the bias current generator comprises a transistor set comprising the one or more second negative threshold transistors; the first negative threshold transistor comprising: a first terminal and a gate terminal, wherein: the first terminal is coupled to a first terminal of the transistor set comprising the one or more second negative threshold transistors; and the gate terminal is coupled to a second terminal of the transistor set comprising the one or more second negative threshold transistors.
8. The apparatus of claim 7, wherein the first negative threshold transistor comprises a second terminal coupled to a first voltage and/or the second terminal of the transistor set, comprising the one or more second negative threshold transistors, is coupled to a second voltage.
9. The apparatus of claim 1, comprising measurement circuitry configured to measure a first quantity that is dependent on the bias current.
10. The apparatus of claim 9, wherein the first quantity comprises a threshold voltage, the threshold voltage being dependent on the bias current.
11. The apparatus of claim 1, wherein the first quantity comprises a threshold voltage, the threshold voltage being dependent on the bias current.
12. The apparatus of claim 1, wherein the trimming circuitry comprises a memory element configured to store a look up table comprising a plurality of trimming signal settings, each trimming signal setting being associated with a value of the first quantity.
13. The apparatus of claim 12, wherein the trimming circuitry is configured to determine the trimming signal by: selecting, from the look up table, the trimming signal setting associated with

the value of the first quantity as received by the trimming circuitry; and setting the trimming signal based on the trimming signal setting as selected, thereby determining the trimming signal.

14. The apparatus of claim 13, wherein the first quantity comprises a threshold voltage, the threshold voltage being dependent on the bias current, such that the value of the first quantity as received by the trimming circuitry circuit is the value of the threshold voltage measured.

15. The apparatus of claim 1, wherein: the two or more negative threshold transistors comprises: a first negative threshold transistor; and a plurality of second negative threshold transistors; the bias current generator comprises a transistor set comprising a plurality of bit portions, each bit portion comprising at least one of the second negative threshold transistors; and the trimming signal is configured to set each bit portion in an enabled or disabled state.

16. The apparatus of claim 15, wherein the trimming circuitry comprises a memory element configured to store a look up table comprising a plurality of trimming signal settings, each trimming signal setting being associated with a value of the first quantity.

17. The apparatus of claim 16, wherein the trimming circuitry is configured to determine the trimming signal by: selecting, from the look up table, the trimming signal setting associated with the value of the first quantity as received by the trimming circuitry; and setting the trimming signal based on the trimming signal setting as selected, thereby determining the trimming signal.

18. A method comprising: generating a bias current by a bias current generator comprising two or more negative threshold transistors coupled in series; measuring, by a measurement circuitry, a first quantity that is dependent on the bias current; based on the measurement of the first quantity, determining, by a trimming circuit, a trimming signal necessary to provide to the bias current generator such that the bias current generator will adjust the bias current to a desired bias current; and providing, by the trimming circuitry, the determined trimming signal to the bias current generator, wherein the bias current generator adjusts the bias current in response to the trimming signal to provide the desired bias current.
